



Cell efficiency has also been on the rise and poly-crystalline cells now have an efficiency of 18 per cent and mono-crystalline almost 23 per cent.

— S.S.N. Prasad, director and in-charge of the cell line, manufacturing and R&D, Solar Semiconductor



mono-crystalline almost 23 per cent. Also, with increasing standardisation of manufacturing equipment and improving efficiencies of modules, it is expected that there will be a reduction in production costs in the medium term," informs S.S.N. Prasad, director and in-charge of the cell line, manufacturing and R&D, Solar Semiconductor.

The near-term target for the industry is to produce cells with improved efficiencies (19 per cent) for crystalline silicon over the next two-three years and achieve solar cell manufacturing cost of less than \$1/Wp within the next two years.

The primary way to achieve reduction in the cost is through rapid reduction in material cost with improved factory utilisations and yields, capacity expansion and leveraging the benefits of economies of scale, and improved efficiencies for crystalline-silicon manufacturing through technology transfer.

Emphasis is on R&D for new technologies (including organic and advanced inorganic materials) that can improve system efficiency and maintain low-cost production.

Market share for crystalline silicon and thin-film solar cells

Globally, the market share of crystalline silicon solar cells is 80-90 per cent compared to 10-20 per cent of thin-film solar cells. In India too, crystalline silicon enjoys a predominant portion (90 per cent) of the market. The installed capacity for thin-film solar cells is just ramping up in India, while there is a relatively large installed base for crystalline silicon solar cells.

"Domination of crystalline technology is primarily because until now most applications of solar technology in India have been small-scale (traffic lights, streetlights, etc)," says Ravi Surapaneni, vice president, Solar Semiconductor. Now, with focus on large-scale solar power plants, we can expect thin-film technology to gain market share in commercial applications. Thin-film offers cost advantage in large-scale applications even with lower efficiency.

Challenges for solar cell production

The main challenge before solar cell manufacturers is to reduce the cost of devices through increase in efficiency, reduction of rejection and increase in volume of production (so that overheads are spread over a large volume).

The manufacturing base in India comprises cell and module manufacturing, with the bulk of the value addition taking place outside the country. Additionally, the current scale of manufacturing in India is small in comparison to global standards. To give you an example, the total manufacturing capacity of the largest solar company in 2009 was 1100 MW, whereas in India the combined capacity of all the producers put together is only around

500 MW at present.

Hence two issues need to be addressed: scale and integration. Significant and immediate steps are required from the government of India to facilitate a bigger and vertically integrated manufacturing base in the country. Ramping up production capacity and levels can lead to increased economies of scale and thus higher availability of cells.

Up to now the size of the Indian market has been very small as it was restricted to the off-grid mode, whereas large volumes can only be generated in MW-scale grid-connected plants. Hence Indian manufacturers have to traditionally compete with global suppliers and find global markets in demanding markets such as Germany, Spain and the US.

At present, in India, there is no manufacturing capacity for polysilicon and silicon wafers—the key raw material for cell manufacturing—and the country is entirely dependent on imports for the same. "Indian companies enter the value chain from wafer onwards and manufacture PV cells from these and then assemble the cells into modules and so on. However, now that the Indian solar market is opening up and promises to grow rapidly, it is likely that India will see fresh investments into the upstream manufacturing as well," says K. Subramanya, CEO, Tata BP Solar.

Solar cell wafer modules can be imported duty-free, whereas consumables required for making these items attract duty. This puts wafer manufacturing in India at a disadvantage. Maharishi Solar claims to be the only company in India to make wafers for



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